Vanshaj P Mohan

Master of Computer Application
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GitHub Profile
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Education

- **Credit Linked Program in Data Science** *2025-2026*
  IIT Guwahati X Masai, Assam
- **Master of Computer Applications** *2024-2026*
  Christ University, Delhi NCR
- **BSc Statistics and Computer Science** *2020-2023*
  Lucknow University, Lucknow

Experience

- **LASA Finance** *Jan 2026 – Mar 2026*
  React Intern Noida
  - Built a real-time financial dashboard using React and TypeScript, featuring dynamic stock screeners and advanced charting for technical analysis.
  - Engineered an automated data pipeline using the Google Sheets API to seamlessly ingest live market data, eliminating manual workflows and improving data accuracy.
- **Digipodium** *May 2025 – July 2025*
  Data Science Intern Lucknow
  - Developed HandSight, a real-time hand tracking application using Python, OpenCV, and MediaPipe.
  - Designed and implemented a computer vision pipeline to detect and visualize 21 hand landmarks from live webcam input, enabling responsive gesture recognition. Optimized performance to achieve 25–35 FPS and built modular code to support future gesture classification.
- **GirlScript Summer of Code** *July 2025 – Sept 2025*
  Open Source Contributor Open Source Program

Personal Projects

- **WaveBrush** *July 2025 - Sept 2025*
  WaveBrush is an AI-powered virtual painter that lets you draw in the air using just your hand gestures.
  - Built with Python, OpenCV, and MediaPipe, this project captures webcam input, tracks hand landmarks in real time, and transforms fingertip movements into smooth brush strokes with dynamic color and erasing controls.
  - Processes webcam input and detects landmarks for accurate finger tracking with Index fingertip acts as the brush pointer adding Gesture logic controls drawing, erasing, and switching colors
  - Gesture logic controls drawing, erasing, and switching colors.
  - **Technologies Used:** Python, OpenCV, MediaPipe, NumPy
- **HandIQ** *July 2025 - July 2025*
  A real-time hand tracking and finger counting application built using Python, MediaPipe, and OpenCV.
  - The idea behind HandIQ was to explore gesture recognition by leveraging landmark-based tracking to identify hand poses and convert them into meaningful visual feedback.
  - Thumb is evaluated based on x-coordinates because of its sideways orientation. The remaining four fingers use y-coordinates to determine whether they're open or closed..
  - Preloaded images (0–5) are displayed as overlays depending on the number of fingers raised. A green rectangle added with numeric count is drawn on the image frame for clarity..
  - Optimized for right-hand use only (left-hand support planned).
  - **Technologies Used:** Python, OpenCV, MediaPipe, NumPy (implicitly via OpenCV)
- **SuperstoreViz** *May 2025 - June 2025*
  A dynamic and interactive data visualization tool built using Streamlit to analyze Superstore sales data.
  - This dashboard allows users to upload their own Excel or CSV files, filter data in real-time, and explore dynamic insights through a variety of visualizations.
  - It enables users to explore trends, segment performance, shipping methods, and regional insights through real-time filters and rich visualizations.

  - This dashboard helps users gain deep, actionable insights from Superstore sales data by allowing them to Explore Sales Trends, Visualize Product Performance, Analyze Profitability and many more.
  - **Technologies Used:** Python, Streamlit, Plotly, Excel/CSV

- **HandSight — Real-Time Hand Tracking** *June 2025 - June 2025*
  This project is a real-time hand tracking application developed using Python, MediaPipe, and OpenCV.
  - It captures live video from your webcam, detects hand landmarks, and overlays them along with frames per second (FPS) in a display window. Ideal for anyone exploring gesture recognition, computer vision, or interactive applications using hand inputs.
  - Real-time hand detection and landmark tracking with Coordinates of each landmark printed to console also Highlights the wrist (landmark ID 0) with a colored circle.
  - On-screen FPS counter to evaluate real-time performance with Landmark coordinates printed live for reference or future model training.
  - **Technologies Used:** Python (OpenCV, MediaPipe), CV2 for frame manipulation and visualization

- **Adidas Interactive Sales Dashboard**
  Built an interactive sales analytics dashboard using Streamlit and Plotly for real-time visualization of Adidas sales data.
  - Integrated **Excel-based sales dataset** into Streamlit for dynamic reporting and analysis.
  - Designed multiple interactive charts including retailer-wise sales, monthly sales trends, state-wise sales with dual axes, and treemaps for region and city sales.
  - Implemented advanced features such as data download options, expanders for detailed insights, and custom styling for professional UI.
  - Delivered additional insights through visualizations like operating profit by city and price per unit distribution by region.
  - **Technologies Used:** Python, Streamlit, Pandas, Plotly, PIL, Excel

Technical Skills and Interests

Languages: Python, C, C++, JavaScript, SQL

Data Science Libraries: Numpy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, TensorFlow

Machine Learning: Data Cleaning, Data Visualization, Exploratory Data Analysis

Databases: MySQL, MongoDB

Tools: Jupyter Notebook, Microsoft Excel, MS Word, MS Power Point, Google Sheets, Power BI, Tableau

Web Technology: MERN Stack

Soft Skills: Team Management, Problem Solving, Presentation, Adaptability

Achievements and Certifications

- IBM AI Fundamentals (September 2025) Certificate
- Introduction to Prompt Engineering with GitHub Copilot (July 2025) Certificate
- Deloitte Cyber Job Simulation (June 2025) Certificate
- Deloitte Data Analytics Job Simulation (June 2025) Certificate
- Gemini for Google Workspace (June 2025) Certificate
- Introduction to Generative AI Studio (June 2025) Certificate
- Introduction to Cybersecurity Awareness Certificate (May 2025) Certificate
- Data Science and Analytics Certificate (Dec 2024) Certificate
- MERN Stack Web Development Certificate (Aug 2024) Certificate
- Python Data Science Project Certificate (April 2024) Certificate
- Python 101 for Data Science (April 2024) Certificate
- Python Data Science Course Completion Certificate (March 2024) Certificate
- Has a total of 4,090 contributions on GitHub.
- Selected as a Contributor for Social Summer of Code (SSOC) Season 4 June'25.